

Statement of Purpose

M.S. Technology Innovation and Entrepreneurship — KAUST

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How can analytical models for smart-city investment become tools that entrepreneurs, infrastructure planners, and public-sector decision makers can use? My academic path has moved from financial evaluation to urban simulation and then to AI-enabled investment scoring. KAUST's M.S. in Technology Innovation and Entrepreneurship is the program where I want to turn that trajectory into a product-development discipline: building models and translating them into venture-ready decision tools for smart-city infrastructure, mobility, and sustainable urban investment.

My first applied research experience came through my final-year project at the Université Internationale de Casablanca, where I studied the financial profitability of outdoor advertising assets in Casablanca. Advertising operators lacked a structured way to decide where to invest in static or digital billboards under traffic, demographic, regulatory, and cost constraints. I built a mixed-methods model using SUMO traffic simulation, GIS mapping, demographic analysis, and CAPEX/OPEX/TCO evaluation. I then developed a Potential Profitability Index to compare locations and support investment decisions. This project taught me that urban data only becomes useful when it becomes a decision process. It also exposed a limitation I want to address at KAUST: I could model profitability, but I still needed stronger skills in product design, user validation, and scalable tool development.

I extended that logic in my AI-MCDM investment scoring paper on smart-city infrastructure under carbon, energy, and geoeconomic risk. The problem shifted from advertising assets to broader infrastructure allocation. I designed a four-layer decision framework combining XGBoost-based demand forecasting, carbon-adjusted valuation, Real Options logic, Fuzzy-AHP/MEREC weighting, and AROMAN ranking. The model reduced mean absolute error by 45.8 percent relative to Ridge regression and showed that multi-criteria rankings diverged from NPV-only rankings across 25 projects. The main lesson was not only technical performance. Complex models must remain auditable, traceable, and explainable if investors or public institutions are expected to trust them. The project also tested my resilience: early implementations produced unstable rankings, and the geoeconomic-risk hypothesis did not hold at the full panel level. I corrected the normalization logic, separated supported findings from exploratory ones, and learned to treat negative evidence as part of the research process.

My internships reinforced the same problem from an operational angle. At SAN MAROC, I segmented a 97.9B MAD non-performing loan portfolio and automated reporting workflows using VBA and Power Query. At 2ACOM, I built financial models for OOH and DOOH scenarios and linked profitability to spatial and traffic variables. At MAGHREBAIL, I worked on corporate credit-risk reporting. These experiences trained me to connect data, financial logic, operational constraints, and managerial decisions. They also showed me why many analytical models fail outside academic settings: they do not reach the form of a usable interface, a validated product, or a clear business case.

At KAUST, I want to develop a venture-oriented decision-support platform for smart-city infrastructure prioritization. The first version would combine three modules: a mobility and demand-forecasting layer, a financial and carbon-risk scoring layer, and an explainable recommendation interface for planners, investors, or startup teams. Professor Amr Abdel-Hamid's work in Technology Innovation and Entrepreneurship, engineering systems, AIoT, edge-oriented applications, and

prototyping aligns with my goal of turning analytical models into testable smart-city tools. Professor Mikkell Draebye's entrepreneurship teaching and focus on innovation processes match the commercialization side of the project: customer discovery, market validation, business-model design, and venture formation. Professor Aamir Farooq's role as TIE Program Director also matters because the program joins technological rigor, product development, and entrepreneurial execution.

The TIE program attracts me because it treats innovation as execution. Its project-based structure, prototyping cycles, commercialization training, and Global Technology Hub internship match the skills I need to acquire. I want to use KAUST's prototyping environment to move from spreadsheet-based models toward a working MVP: a dashboard or scoring engine that can be tested with users, refined through feedback, and positioned for smart-city or infrastructure-finance markets. I am ready to contribute through literature mapping, data cleaning, financial modelling, Python-based prototyping, validation experiments, and documentation. I am also ready to learn from engineers, founders, and product mentors.

My short-term goal after KAUST is to work at the intersection of smart-city technology, infrastructure analytics, and venture development, either in a startup, innovation lab, or corporate venture environment. My longer-term goal is to build decision-support products that help emerging cities allocate capital more transparently under climate, energy, mobility, and geopolitical uncertainty. I am applying to KAUST TIE because the program matches the researcher and builder I am trying to become: someone who can identify a real urban problem, model it rigorously, prototype a usable solution, and test whether it deserves to become a venture.